CCC-C-419E
April 27, 1977
SUPERSEDING
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September 28, 1973

FEDERAL SPECIFICATION

CLOTH, DUCK, COTTON, UNBLEACHED, PLIED-YARNS

ARMY AND NUMBERED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for grey (unbleached) cotton duck cloth.

1.2 Classification.

1.2.1 Types. The cotton duck cloth shall be of the following types as specified (see 6.2):

Type I - Numbered duck.
Type III - Army duck.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

V-T-276 - Thread, Cotton.
PPP-P-1135 - Packaging and Packing of Duck Fabrics (Cotton; Synthetic Fiber; Cotton-Synthetic Fiber Blends).

Federal Standards:

FED-STD-4 - Glossary of Fabric Imperfections.
FED-STD-191 - Textile Test Methods.
FED-STD-751 - Stitches, Seams, and Stitching.

FSC 8305

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(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

Rules and Regulations Under the Textile Fiber Products Identification Act.

(Copies may be obtained without charge from the Federal Trade Commission, Washington, DC 20580.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Yarn. The yarn shall be cotton which has been carded, drawn, and spun into singles or plied yarns as specified in tables I and II.

3.2 Physical requirements. The cloth shall be as specified (see 6.2) and shall conform to the requirements specified in tables I and II when tested as specified in 4.4.



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TABLE I. Physical requirements for type I, numbered duck

| Number of duck, commercial designation | Weight oz per sq yd ($g/m^2$) minimum | Breaking strength pound (Newton) minimum | | Air permeability $ft^3/(min.\ ft^2)$ $cm^3/(s.cm^2)$ maximum |
|---|---|---|---|---|
| | | Warp | Filling | |
| Hard texture: | | | | |
| 1 1/ | 28.71 (973) | 440 (1958) | 370 (1647) | 2 (1.02) |
| 2 1/ | 27.12 (919) | 420 (1869) | 345 (1535) | 2 (1.02) |
| 4 1/ | 23.93 (811) | 375 (1669) | 300 (1335) | 2 (1.02) |
| 6 1/ | 20.74 (703) | 335 (1491) | 250 (1113) | 2 (1.02) |
| 8 1/ | 17.55 (595) | 285 (1268) | 210 (935) | 2 (1.02) |
| 10 2/ | 14.35 (486) | 245 (1090) | 160 (712) | 4 (2.03) |
| 12 2/ | 11.16 (378) | 195 (868) | 120 (534) | 4 (2.03) |
| Medium texture: | | | | |
| 1 1/ | 28.71 (973) | 425 (1891) | 345 (1535) | 4 (2.03) |
| 4 1/ | 23.93 (811) | 350 (1558) | 290 (1291) | 4 (2.03) |
| 6 1/ | 20.74 (703) | 305 (1357) | 250 (1113) | 4 (2.03) |

1/ Plied yarns shall be used for both warp and filling.

2/ Warp yarns shall be plied and the filling yarns shall be singles or plied.





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TABLE II. Physical requirements for type III, Army Duck

| Weight oz per sq yd ($g/m^2$) minimum | Breaking strength pounds (Newton) minimum | | Tearing strength pounds (Newton) minimum | | Air permeability $ft^3/(min.ft^2)$ $cm^3/(s.cm^2)$ maximum |
|---|---|---|---|---|---|
| | Warp | Filling | Warp | Filling | |
| 8.25 (280) 1/ | 125 (556) | 120 (534) | 4.0 (18) | 3.0 (13) | 4.0 (2.03) |
| 9.85 (334) 1/ | 160 (712) | 110 (490) | - - | - - | 4.0 (2.03) |
| 12.29 (417) 1/ | 210 (935) | 130 (579) | - - | - - | 4.0 (2.03) |
| 14.77 (501) 1/ | 235 (1046) | 175 (779) | - - | - - | - - |
| 18.48 (626) 1/ | 315 (1402) | 200 (890) | - - | - - | - - |

1/ The warp yarns shall be plied and the filling yarns shall be singles or plied.



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3.2.1 Width. The width of the cloth inclusive of selvages shall be as specified (see 6.2). The following tolerances shall be permitted:

| Width of duck in inches (cm) | Tolerance in inches (cm) | |
|---|---|---|
| 40 (102 cm) or less | +1/4 (0.6 cm) | -1/4 (0.6 cm) |
| Over 40 (102 cm) to 51 (129 cm) inclusive | +3/8 ( 1 cm) | -1/4 (0.6 cm) |
| Over 51 (129 cm) to 81 (206 cm) inclusive | +5/8 (1.6 cm) | -3/8 ( 1 cm) |
| Over 81 (206 cm) | +3/4 (1.9 cm) | -3/8 ( 1 cm) |

3.2.2 Weave. The weave shall be plain, one up, one down.

3.3 Nonfibrous material content. The cloth shall contain not more than 2.5 percent starch and protein content excluding chloroform-soluble and water-soluble material. The total sizing, finishing and other nonfibrous material content of the finished cloth shall not exceed 5.0 percent when tested as specified in 4.4.

3.4 Length and put up. Unless otherwise specified (see 6.2), the cloth shall be in one continuous piece of not less than 40 yards (36.6 m) and shall be put up in rolls in accordance with PPP-P-1135. The maximum length of the rolls when converted to pounds shall not exceed the maximum weight specified in PPP-P-1135. For duck No. 10 and lighter in weight, pieces shall be sewn together with a butted FSa-1 type seam of FED-STD-751 using ticket No. 10, 3 ply unbleached cotton thread conforming to type IIIA of V-T-276, and having a minimum of 6 stitches per inch (2 stitches per cm). The stitching shall be placed 1/2 to 5/8 (1.3 to 1.6 cm) from the cut or raw edge of the cloth. For cloth heavier in weight than No. 10, the pieces shall be lapped and not sewn together. The number of pieces in the roll shall be indicated by stencil marking on the last 6 inches (15.2 cm) of the roll on the outer surface.

3.5 Fiber identification. Each roll shall be labeled, or ticketed, for fiber content in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

3.6 Face marking. The face side of the cloth shall be identified by applying a stamping on that side of the cloth with the word "Face" at each end of the piece.

3.7 Workmanship. The cloth shall conform to the quality established by this specification. The demerit points per 100 square yards (83.6 $m^2$) when calculated as specified in section 4 shall not exceed the established maximum point value.



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Quality conformance inspection. Sampling for inspection shall be performed in accordance with 4.3 and 4.4. When applicable, definitions of quality assurance terms and procedures shall be in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3 Examination of end item. Examination of the end item shall be in accordance with 4.3.1 through 4.3.3.

4.3.1 Yard-by-yard examination. Each roll in the sample shall be examined on the face side only. When the total yardage in the roll does not exceed 100 yards (91 m), the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards (91 m), only 100 yards (91 m) shall be examined. All defects, as defined in Section I of FED-STD-4 which are clearly noticeable at normal inspection distance 3 feet (1 m) shall be scored and assigned demerit points as listed in 4.3.1.1 with the following exceptions: Glossary numbers 1, 2, 3, 5, 14, 26, 27, and 29 shall not be scored unless they exceed three times the thickness of the yarn. Misdraws and reed marks shall be scored if they result in a clearly noticeable separation of warp yarns. Each knot or slub shall be scored as a defect only when it exceeds three times the normal yarn diameter. No linear yard (increments of 1 yard (0.91 m) on the measuring device of the inspection machine) from any one roll within the sample shall be penalized more than 4 points. The sample size shall be in accordance with the following:



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| Lot size (yards) (0.91 meter) | Sample size (rolls) |
|---|---|
| 3200 or less | 8 |
| 3201 up to including 10,000 | 13 |
| 10,001 and over | 20 |

The lot shall be unacceptable if the points per 100 square yards (83.6 $m^2$) of the total yardage examined exceeds 40 points. The lot shall be unacceptable if the points per 100 square yards (83.6 $m^2$) of two or more individual rolls exceeds 60 points. If one roll exceeds 60 points per 100 square yards (83.6 $m^2$ a second sample of 20 rolls shall be examined only for individual roll quality examination. The lot shall be unacceptable if one or more rolls in the second sample exceed 60 points per 100 square yards (83.6 $m^2$). Point computation for lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample x 3600}}{\text{Contracted width of cloth (inches) x total yards inspected}} = \text{Points per 100 square yards (83.6 } m^2\text{)}$$

4.3.1.1 Demerit points. Demerit points shall be assigned as follows:

For defects up to and including 3 inches (7.6 cm) in any dimension - one point

For defects exceeding 3 inches (7.6 cm) but not exceeding 6 inches (15.2 cm) in any dimension - two points

For defects exceeding 6 inches (15.2 cm) but not exceeding 9 inches (22.9 cm) in any dimension - three points

For defects exceeding 9 inches (22.9 cm) in any dimension - four points

The following defects when present shall be scored four points for each yard (0.91 m) in which they occur:

Baggy, ridgy, or wavy cloth.
Width not within specified tolerance.
Objectionable odor.
Excessive neppiness.

4.3.2 Examination for length.

4.3.2.1 Individual rolls. Each roll in the sample shall be examined for put up and length. Each defect shall be counted not more than once in each roll examined. The lot shall be unacceptable if two or more of the following defects are present.



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Defects

Any length found to be more than 2 yards (1.82 m) less than the length marked on the roll ticket.
Any piece less than 40 yards in length.
For No. 10 duck and lighter in weight:
a. Pieces not sewn together with a butted FSa-1 type seam.
b. Stitches less than the minimum specified.
c. Stitching not 1/2 to 5/8 inch (1.3 to 1.6 cm) from the cut or raw edge of the cloth.
For duck heavier in weight than No. 10, pieces sewn instead of lapped.
Number of pieces in roll not stenciled on last 6 inches (15.2 cm) of rolls on outer surface.

4.3.2.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the tickets.

4.3.3 Examination for face markings and compliance with the Textile Fiber Products Identification Act. During the yard-by-yard examination, each roll in the sample shall be examined for these defects. The lot shall be unacceptable if two or more rolls in the sample have the face stamping missing from either or both ends, or are not labeled or ticketed in accordance with the Textile Fiber Products Identification Act.

4.3.4 Examination of preparation for delivery requirements. An examination shall be made in accordance with the provisions of PPP-P-1135 to determine that packaging, packing and marking comply with section 5 requirements.

4.4 Testing of the end item. The methods of testing specified in FED-STD-191 wherever applicable as listed in table III shall be followed. The physical and chemical values specified in section 3 apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test methods. The sample unit for testing shall be 1-1/2 yards (1.37 m) full width of the cloth. The lot shall be unacceptable if one or more sample units fail to meet any requirements specified. All test reports shall contain the individual values utilized in expressing the final results. The sample size shall be as follows:

| Lot size in yards (0.91 meter) | Sample size |
|---|---|
| 800 or less | 2 |
| 801 up to and including 22,000 | 3 |
| 22,001 and over | 5 |



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TABLE III. Test methods

| Characteristic | Requirement paragraph | Test method |
|---|---|---|
| Yarn ply | 3.2 | Visual 1/ |
| Weight | 3.2 | 5041 |
| Breaking strength: | | |
| Warp | 3.2 | 5100 |
| Filling | 3.2 | 5100 |
| Tearing strength (8.25 oz): | | |
| Warp | 3.2 | 5132 |
| Filling | 3.2 | 5132 |
| Air permeability | 3.2 | 5450 |
| Weave | 3.2.2 | Visual 1/ |
| Starch and protein content excluding chloroform-soluble and water-soluble material | 3.3 | 2611 2/ |
| Total sizing, finishing and other nonfibrous materials | 3.3 | 2611 2/ |

1/ One determination shall be made from each sample unit and the results reported as "pass" or "fail".

2/ A certificate of compliance shall be submitted and will be acceptable for the stated requirement.

5. PREPARATION FOR DELIVERY

5.1 Put-up and packaging. Put-up and packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C. The cloth shall be put-up and packaged in accordance with the applicable requirements of PPP-P-1135.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Levels A, B and C. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1135.



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5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the requirements of PPP-P-1135.

6. NOTES

6.1 Intended use. The cloths are primarily intended for use in fabrication of tentage and equipage items.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) Type, number and weight of cloth required (see 1.2 and 3.2).
(c) Width required (see 3.2.1).
(d) Length, if other than specified (see 3.4).
(e) Selection of applicable levels of put up, packaging and packing (see 5.1 and 5.2).

6.3 Supersession data. Under type I, hard textured duck Nos. 12/0, 8/0, 2/0, 1/0, 3, 5, 11, medium textured duck Nos. 2, 3, 5 and Army Duck 15.90 ounces per square yard have been deleted from this specification. These ducks are no longer required.

Custodians:

Army - GL
Navy - SA
Air Force - 11

Review activities:

Army - MD, MI, MU, EA, EL
Navy - MC
Air Force - 99
DLA - CT

User activities:

Army - AT, ME
Navy - YD

Preparing activity:

Army - GL

Civil Agency Coordinating Activities:

GSA-FSS
HEW

Project No. 8305-0453

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